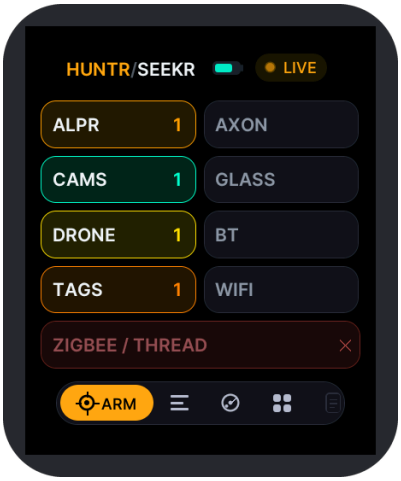
HUNTR/SEEKR

Counter-surveillance signal scanner

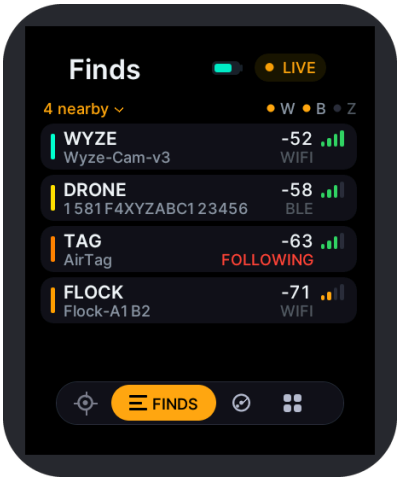
WATCH Edition
User Guide and Start-up Guide

Waveshare ESP32-C6-Touch-AMOLED-2.06

Public Beta 1



Arm groups



See what it finds



Track a signal

Printable guide. Screen pictures are example renderings of the interface; your screen will match the layout.

Contents

1. About this device 3

2. What you need 4

3. Quick start 4

4. Getting around 6

5. The detection groups 8

6. Everyday use 12

7. Drone Remote ID 24

8. Sharing over Wi-Fi with your phone 25

9. Logs, signatures, clock and updates 26

10. Settings reference 30

11. Self-test and troubleshooting 32

12. Care, safety and responsible use 35

Quick-reference card 36

Glossary 37

Appendix A. Technical summary 38

1. About this device

HUNTR/SEEKR turns the Waveshare ESP32-C6 Touch AMOLED 2.06 board into a pocket radio scanner. It listens on the 2.4 GHz band for the wireless signals that certain kinds of devices give off, compares them with built-in lists of known makers and names, and shows you what it heard, how strong each signal is, and lets you follow a signal to its source.

What it can find

- **Wi-Fi devices** such as security cameras, doorbells, smart glasses, drones and license-plate readers, from their network names and hardware addresses. Smart bulbs in set-up mode are tagged LIGHT. It also lists devices asking for a named network (PROBE) and warns of rogue Wi-Fi signs: a de-authentication flood (DEAUTH), an open copy of a protected network name (TWIN) and a transmitter that answers to every name (KARMA) (section 5.2).
- **Bluetooth devices** such as trackers (AirTag, Tile, SmartTag, Google Find Hub), body cameras, smart glasses and drones that announce themselves.
- **Bluetooth types**: beacons (iBeacon, Eddystone, AltBeacon), wearables (watches, fitness bands, rings), smart lights, Fast Pair accessories and Bluetooth Mesh nodes, each shown with a type tag.
- **Zigbee and Thread** smart-home radios (802.15.4), which are separate from Wi-Fi.
- **Drone Remote ID**: many drones broadcast their own serial number and position, which this device can read and display.
- **Possible card skimmers**: Bluetooth LE modules still wearing a factory name such as HC-08 or linvor, the kind sometimes hidden in gas pumps and ATM card readers. See section 6.9 for what this can and cannot do.

How it works in one minute

Every Wi-Fi access point and camera regularly broadcasts a short **beacon** that says who it is: a network name (**SSID**) and a hardware address (**MAC address**). The first three bytes of a MAC address are the **OUI**, a code that identifies the manufacturer. Bluetooth devices do something similar: they **advertise** a name and manufacturer data. HUNTR/SEEKR receives these broadcasts, matches names and OUIs against its lists, and groups the matches into **groups** such as CAMS or TAGS.

It also measures **signal strength (RSSI)** in dBm. A stronger signal usually means the device is closer, which is what lets you track one down.

> **IMPORTANT Read this once**
> Results are hints, not proof. Device names and addresses can be changed or randomized, many devices are not in the lists, and a quiet device may simply be out of range. "Nothing found" never proves nothing is there. Do not treat a match as evidence about a person or place.

> **NOTE About Public Beta 1**
> This is a **public beta**: the first release for general use. It does what this guide describes, but a beta can have rough edges and some behaviour may change in later releases. Everything that depends on the radio environment (what is heard, how strongly) varies from place to place. If something does not behave as described, section 11 lists what to check, and the project page explains how to report a problem.

2. What you need

Item	Notes
Waveshare ESP32-C6-Touch-AMOLED-2.06 board	The board this guide covers. It has the touch AMOLED screen, speaker, motion sensor and clock built in.
USB-C data cable	Must be a data cable, not a charge-only cable. Some cables that came with power banks are charge-only.
A computer	Windows, macOS or Linux, to load the software onto the board.
The firmware file	**HUNTR-SEEKR_Watch_Public_Beta_1.bin** and **SHA256SUMS.txt**, from the project's Releases page (section 3.1).
A flashing tool	**esptool**, free, needs Python 3. Section 3.2 shows the one command you need.
A phone (optional)	Used for the Live view, the clock sync and firmware updates over the board's own Wi-Fi hotspot.
A GPS module (optional)	Connects to the board's UART pads and adds your position to the log.
A lithium battery (optional)	The board can run from USB power alone. If you add a battery, use only the type Waveshare specifies for this board.

> **TIP One device**
> This guide is for the **WATCH** (Waveshare ESP32-C6-Touch-AMOLED-2.06). The boot screen shows the software version in small text at the bottom.

3. Quick start

3.1 Get the firmware file

Download **HUNTR-SEEKR_Watch_Public_Beta_1.bin** and **SHA256SUMS.txt** from the project's Releases page and keep them in one folder. To check the download is intact, compare its checksum with the line in SHA256SUMS.txt (optional but recommended):

Computer	Command
Windows (PowerShell)	**Get-FileHash .\HUNTR-SEEKR_Watch_Public_Beta_1.bin -Algorithm SHA256**
macOS or Linux	**shasum -a 256 -c SHA256SUMS.txt**

3.2 Flash it to the watch

1. Install **Python 3** (python.org) if you do not have it, then install the flashing tool once: **pip install esptool** (on Windows, if pip is not found: **py -m pip install esptool**).
2. Plug the watch into the computer with the USB-C data cable.
3. Find the port: on **Windows** it is a COM port in Device Manager (for example COM5); on **macOS** it looks like /dev/cu.usbmodem... (list with ls /dev/cu.*); on **Linux** it is usually /dev/ttyACM0.

4 In a terminal, in the folder with the file, run (replace YOUR_PORT):
esptool --chip esp32c6 --port YOUR_PORT --baud 460800 write-flash 0x0 HUNTR-SEEKR_Watch_Public_Beta_1.bin

5 Wait for **Hash of data verified** and **Hard resetting**. The watch restarts by itself.

> **TIP If the computer cannot connect to the watch**
> Use a USB-C **data** cable, and try another port on the computer. Then put the chip in download mode: unplug the cable, hold the watch's **BOOT** side button, plug the cable in, release BOOT, and run the command again. If the connection is unreliable, use **--baud 115200**.

> **NOTE What flashing changes**
> Writing the file replaces the software and **resets the saved settings**. Logs that were saved on the watch are kept. To update later you can flash the newer file the same way, or update from your phone without a computer (section 9.4).

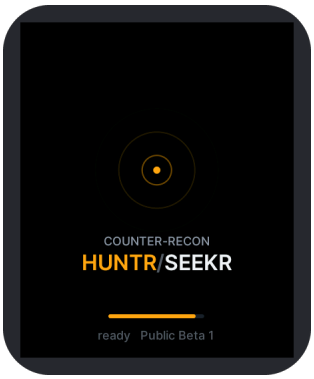
3.3 First start-up checklist

1 The screen shows a small radar animation and the name, then a line at the bottom: **ready Public Beta 1**.

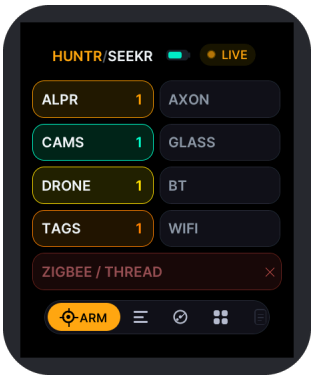
2 A few seconds later the **ARM** page appears with nine group tiles. That is the home screen.

3 If the bottom line says **touch NOT detected** in red, see Troubleshooting (section 11.2).

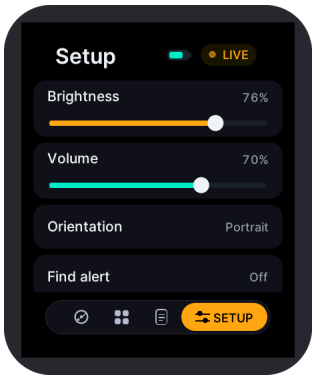
4 Tap the tiles to check that touch works. Then go to **SETUP > Self-test** (see section 11). Every dot should be green. A row that is grey is a part your watch does not have, which is normal. **Clock** and **Battery** can be amber until a clock is set or a battery is fitted.



Boot screen



ARM page (home)



SETUP page

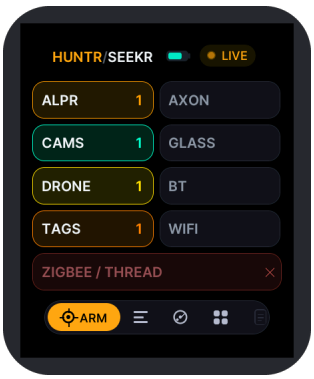
4. Getting around

The watch has six pages. Swipe left and right to move between them, or tap an icon on the bar along the bottom. The page you are on is the amber pill with its name.

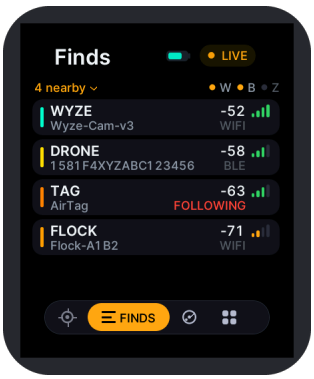
Page	What it is for
ARM	Turn detection groups on and off. Each group is a tile.
FINDS	The live list of everything found (strongest first by default; tap the count to change the view).
SWEEP	Walkthrough marks (which room was each device strongest in), a baseline of what is normally in a place, and a list of what is new. See section 6.13.
GROUPS	Every group with what it looks for, its live results, and one-tap Scenes.
LOG	The findings record (SAVE LOG), counters, battery, the Share button and the Session Report.
SETUP	Brightness, volume, orientation, alerts and other settings.

The icon bar

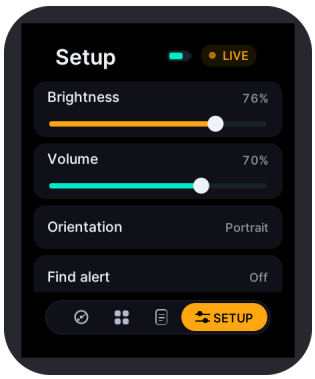
Along the bottom is a strip of page icons. The page you are on is the **amber pill** with its name, kept in the middle of the bar. As you swipe from page to page, **the strip slides along with your finger**. Icons near the ends of the bar fade out and are cut off: that means **there are more pages that way**. Tap any icon you can see to go straight to that page.



The first page: ARM



The next page: the strip slides



The last page: SETUP

Touch gestures

Do this	To get this
Tap	Select or press a button, arm or disarm a tile, choose a row.
Swipe left / right	Move to the next / previous page.
Drag up / down	Scroll a long list or page.
Long-press a tile (about half a second)	Open the list of finds for just that group.
Long-press a find in a list (about one second)	Hide that device. Bring it back from the Ignored screen (tap N hidden).
Tap the status pill (top right)	Pause or resume scanning (or give three hard shakes if Shake to pause is on).
Long-press the status pill	Switch Discreet mode on or off.
Press BOOT briefly	Show the watch face; press again to go back (section 6.11).
Keep swiping left on the last page (SETUP)	The Power screen (section 6.10).
Touch a sleeping screen	The first touch only wakes the screen. Touch again to act.

The status pill and battery icon

The rounded label at the top right tells you what the watch is doing. The small battery icon beside it is green normally, amber under 35% and red under 15%, and teal while charging.

Pill says	Meaning
LIVE	Scanning is running normally.
PAUSED	You (or a shake) paused scanning. Nothing new is recorded.
IDLE	No group is armed, so nothing is being scanned.
HUNT	You are tracking one device.
QUIET	Discreet mode is on: silent, dim, no banners.
SAVER	Power saver is active: dimmer, shorter screen time, slower scanning.
SKIM?	A possible card skimmer was heard in the last 3 minutes (section 6.9). The pill is red.

5. The detection groups

Everything the device looks for is organized into nine groups. Arm the ones you care about; the device only records matches for armed groups.

Group	Looks for	Radio
ALPR	Flock (cameras, Raven sensors, battery packs), Penguin, Pigvision, Ubicquia and Genetec AutoVu (license-plate readers)	Wi-Fi and Bluetooth
AXON	Body cameras, docks, TASER (Axon equipment), also recognised by their Bluetooth service data and company ID	Wi-Fi and Bluetooth
CAMS	Wyze, Ring, Tapo, Reolink, Eufy, Nest, Arlo, Blink, Amcrest, Lorex, Hikvision, Dahua, Axis, Uniview, Vivotek, Hanwha cameras, and the names low-cost hidden-camera modules use; police body and dash cameras (Digital Ally, WatchGuard, i-PRO, Utility BodyWorn, Motorola VideoManager); Verkada, Avigilon	Wi-Fi and Bluetooth
GLASS	Ray-Ban Meta (Wayfarer, Oakley Meta), Spectacles, Vuzix smart glasses	Wi-Fi and Bluetooth
DRONE	DJI, Autel, Skydio, Parrot, and Remote ID broadcasts	Wi-Fi and Bluetooth
BT	Bluetooth devices by name or maker, with type tags (see section 5.1): IBEACON, EDDYSTONE, ALTBEACON, WEARABLE, LIGHT, FASTPAIR, MESH; also named when recognised: public-safety radios and vehicle light-bar controllers (Motorola, L3Harris, Tait, Feniex, SoundOff, Whelen)	Bluetooth
TAGS	Tile, AirTag and other Find My trackers, SmartTag, Google Find Hub trackers, Chipolo, Pebblebee, TrackR, Jiobit	Bluetooth
WIFI	Any nearby 2.4 GHz Wi-Fi network (access point), and devices with a fixed address asking for a named network (tag PROBE). Rogue Wi-Fi signs are tagged DEAUTH, TWIN and KARMA (section 5.2). Smart bulbs in set-up mode (WiZ, LIFX, Yeelight) are tagged LIGHT. Vehicle routers (Cradlepoint, Sierra Wireless, Peplink), rugged laptops and handhelds (Panasonic, Zebra, Datalogic) are named when recognised	Wi-Fi
ZB/TH	Zigbee, Thread and plain 802.15.4 devices (channels 11 to 26)	802.15.4

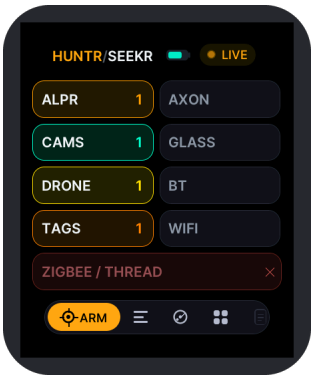
> **NOTE About the public-safety signatures**
> Many of these signatures come from a published list and are marked there as **possible**: the maker also sells ordinary products, so a match is a hint, never proof. Those are listed quietly (low confidence, no alert). What it cannot see: radios and accessories that use only **Bluetooth Classic** (many Motorola, Harris, Tait and Kenwood models), anything on 5 GHz Wi-Fi (this chip is 2.4 GHz only), and devices that use a random address. Agency fleet Wi-Fi names are usually changed from the maker's default, so a router can go unseen. The device recognises 166 of the 184 signatures in the list it was built from by itself (no rules file needed); the other 18 need abilities these chips lack, such as Bluetooth Classic.

The one rule: Wi-Fi groups and ZB/TH cannot run together

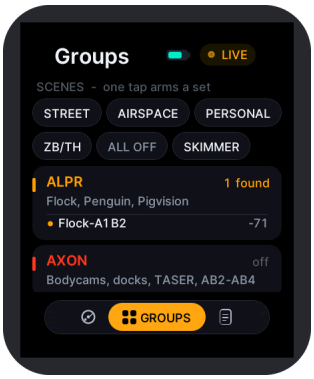
The board has a single 2.4 GHz radio that is shared between Wi-Fi, Bluetooth and 802.15.4. Bluetooth groups can run alongside either, but Wi-Fi scanning and Zigbee/Thread scanning both need the radio to itself, so the device lets you arm **one side at a time**:

- **Wi-Fi side:** ALPR, AXON, CAMS, GLASS, DRONE, WIFI.
- **802.15.4 side:** ZB/TH only.
- **BT** and **TAGS** work with either side.

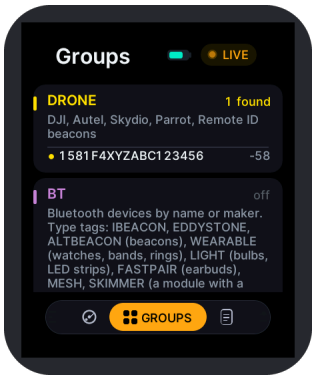
When one side is armed, every tile on the other side turns dim red and says **blocked**. Tapping a blocked tile shakes it and shows a short message telling you what to switch off first.



ARM page: the red ZB/TH tile is blocked while Wi-Fi groups are armed



GROUPS page with Scenes



A group card lists its strongest targets

Signal strength (dBm) in plain terms

Signal strength is shown as a negative number in dBm. **Closer to zero is stronger.** Each 10 dB is roughly a tenfold change in power.

Reading	What it usually means
-30 to -50	Very close: within a few metres, often the same room.
-50 to -65	Close: same room or just through a wall.
-65 to -80	Moderate: another room, or 10 to 20 metres in the open.
-80 to -90	Weak: far away or heavily blocked.
below -90	Barely detectable.

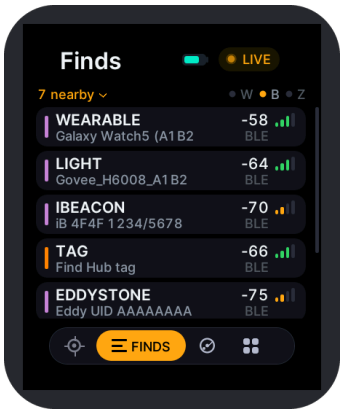
The bars beside a reading: 4 bars is stronger than -55, 3 is stronger than -65, 2 is stronger than -75, 1 is stronger than -88. Treat all of this as rough. Walls, your own body and the other device's antenna can change a reading by 10 to 20 dB.

5.1 Bluetooth type tags

Devices found by the **BT** group carry a tag in the list that says what kind of device they appear to be, instead of a plain "BT". Unnamed devices are shown by their maker (for example "Garmin" or "Huami") using the official Bluetooth company list, not a bare number.

Tag	What it means	How it is recognized
IBEACON	Apple iBeacon (shops, museums, asset tags). The row shows a short ID and the major/minor numbers, e.g. iB 4F52 1234/5678.	Apple's beacon message inside the Bluetooth advertisement.
EDDYSTONE	Google's open beacon format. Shows the beacon ID or the web address it broadcasts.	The Eddystone service and frame type (UID, URL, TLM, EID).
ALTBEACON	An open beacon format used by many vendors.	The 0xBEAC marker in the manufacturer data.
WEARABLE	Watches, fitness bands, heart-rate straps and rings (Garmin, Amazfit/Huami, Polar, Suunto, Withings, Oura, Galaxy Watch and similar).	The maker, the device name, or the device's own "I am a watch" appearance code.
LIGHT	Bluetooth bulbs and LED strips (Govee, Philips Hue, Yeelight, Nanoleaf, common strip controllers).	The maker, the name, or the Hue service. Wi-Fi bulbs in set-up mode are tagged the same way.
FASTPAIR	An Android Fast Pair accessory, typically earbuds or headphones. Not a tracker.	Google's Fast Pair service.
MESH	A Bluetooth Mesh node (often lighting or sensors).	The Mesh proxy or provisioning service.

NOTE Tags are best guesses
A device can call itself anything, and many phones and watches hide behind changing addresses and say almost nothing. An Apple Watch or iPhone usually appears only as "Apple", or not at all. Zigbee bulbs (Hue bridge, IKEA) can be seen only as a Zigbee network under ZB/TH, never by name, because their radio traffic is encrypted. Type-tagged finds are quiet: they are listed but do not trigger chimes, banners or log entries.



FINDS list with type tags:
WEARABLE, LIGHT, IBEACON

5.2 Bluetooth share, rogue Wi-Fi signs and other alerts

While a Wi-Fi group is scanning, the watch also watches for three signs of rogue Wi-Fi behaviour, and lists devices that are asking for a named network. It listens on 2.4 GHz only; this chip does not hear 5 GHz Wi-Fi.

SETUP > Bluetooth share (Off by default) is for when Wi-Fi and Bluetooth groups are armed together. With it On, Wi-Fi scanning pauses for 2 seconds in every 6 so Bluetooth gets the whole radio for a while. Turn it On if Bluetooth finds are slow to appear while a Wi-Fi group is armed. The board has one radio that Wi-Fi and

Bluetooth share.

Switching groups on and off. Arm and disarm groups in any order and as often as you like. When only Bluetooth groups are armed, the Wi-Fi radio is switched off so Bluetooth has the board to itself, and it starts again as soon as a Wi-Fi group is armed. While Wi-Fi and Bluetooth are both running, Bluetooth listens quietly and every 25 seconds asks nearby devices for their names for a few seconds.

Tag	What it means	How it is decided
DEAUTH	A transmitter is flooding de-authentication frames, which knock devices off a Wi-Fi network. It can be a Wi-Fi attack or a faulty device.	15 or more de-authentication or disassociation frames from one address within one second.
TWIN	A network name that is also heard on a password-protected network from a different maker, but this copy is open. It may be an "evil twin" set up to catch people who join.	The same name on an open access point and on a protected one whose hardware address belongs to a different maker.
KARMA	One transmitter that answers to many different network names, which is how some attack gadgets lure phones.	Replies for 5 or more different names within 30 seconds.
PROBE	A device with a fixed (not randomized) address is asking for a particular network name, which is shown. Listed in the WIFI group at low confidence.	Probe requests only. Phones that randomize their address are not listed.

> **NOTE Warnings, not proof**
> A venue with several access points, a hotspot or a repeater can look like a TWIN, and a busy network can look like a DEAUTH. DEAUTH, TWIN and KARMA are announced (chime and banner, following the Find alert setting) whichever groups are armed, as long as at least one Wi-Fi group is armed so that scanning is running. Turn them off with **SETUP > Rogue Wi-Fi alert**.

Other things it notices

- **Device fingerprint.** Many Wi-Fi devices can be told apart by the way they describe themselves when they ask to join a network. The watch adds a short code (for example **IE-fp 3A91C0F2**) to the name of a Wi-Fi device it has heard, so the same hardware can be recognised again even if its name or address changes. You can turn a code into a rule (section 9.3).
- **Drone Remote ID over Wi-Fi.** Drones that send Remote ID using Wi-Fi Aware (NAN) are read and shown in the DRONE group, the same as Bluetooth and beacon Remote ID.
- **Bluetooth spam.** Phones and gadgets that flood the air with fake pairing pop-ups (Apple, Fast Pair and Swift Pair) are recognised and shown once as **BLE SPAM**, so they do not fill the list with hundreds of fake devices.
- **Hidden-camera module names.** Names that low-cost hidden-camera modules often use (for example IPC-123456, HDWiFiCam, YsxLite, A9-xxxx or P2P-xxxx) are listed under CAMS at low confidence. Many harmless products use similar names, so treat one as a reason to look closer, not as a finding.

6. Everyday use

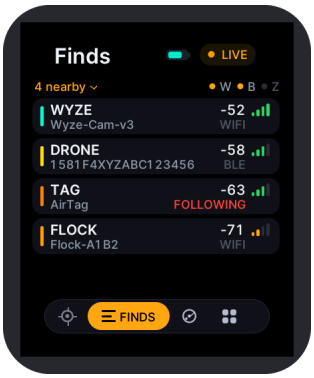
6.1 Arm groups and use Scenes

1. Go to the **ARM** page and tap a tile. It lights up and the pill changes to LIVE. Tap it again to disarm.
2. A number on the tile counts the finds in that group. The small bars on an armed tile show the strongest signal heard in that group in the last 10 seconds.
3. For a quick setup, open **GROUPS** and tap a Scene chip at the top.

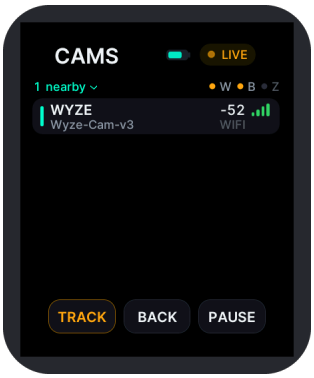
Scene	Arms
STREET	ALPR, AXON, CAMS, GLASS
AIRSPACE	DRONE
PERSONAL	TAGS and GLASS
ZB/TH	Zigbee/Thread only (turns Wi-Fi groups off)
ALL OFF	Disarms everything
SKIMMER	Bluetooth only, and switches the Skimmer alert on (section 6.9)

6.2 Read the list of finds

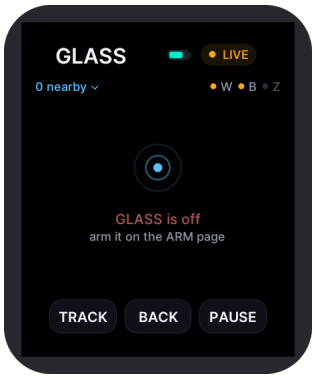
FINDS lists everything found, strongest first. Each row shows a colored group bar, the type, the device name or ID, its signal in dBm and bars, and how it was seen (WIFI, BLE or 802.15.4). A red **FOLLOWING** tag marks a tracker that has been near you for a long time (section 6.6). Long-press a tile on the ARM page, or tap a group on the GROUPS page, to see only that group.



FINDS list



One group's list



Empty list explains why

Choosing one device in a busy area

When the list is long it is easy to tap the wrong row, so the list is built to hold still and to be narrowed:

- **It does not move while you reach for a row.** The order is frozen while a finger is on the list, for three seconds after, and while the list is scrolled down. New finds join at the end until it settles, then the list re-sorts by signal.

- **Tap the count above the list** (for example "12 nearby v") to change the view: **Strongest** (default), **Strong only** (hides anything weaker than -75 dBm and says how many it hid), **Newest** (latest find first) or **A-Z** (by name). The same choice is in **SETUP > List view** and is remembered.
- A device you have selected is never filtered out, so you can always track it. If *Strong only* hides everything, the list says so and tells you to tap the count.
- **Tap a row to select it, then tap it again** (or tap TRACK) to start tracking. Long-press to hide one (section 6.4).

6.3 Track a device

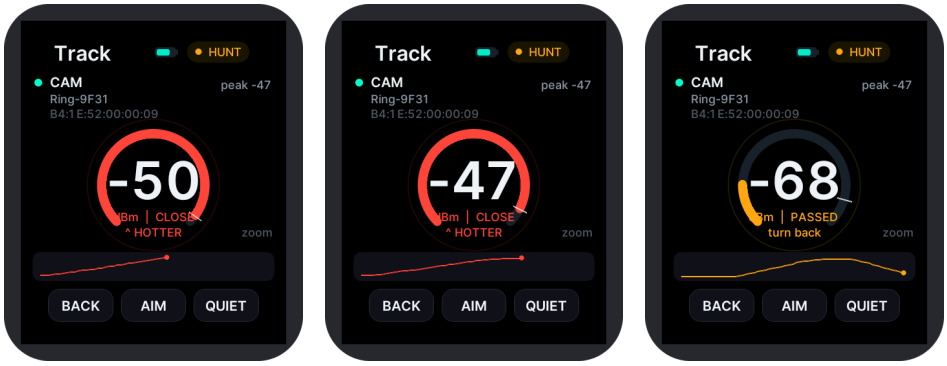
1. On **FINDS** tap the device row, then tap the amber **TRACK** button. (You can also tap a banner when a find is announced.)
2. The tracking screen shows the live signal as a big number, an arc that fills as the signal gets stronger, and a graph of the last few seconds.
3. Walk slowly. **Stronger (closer to zero) means warmer.** The word under the number changes: FAR, NEAR, then CLOSE (stronger than -55).
4. The speaker beeps faster and higher as you get closer. Tap **QUIET** to mute the beeps, **SOUND** to bring them back, and **BACK** to leave. Tap **AIM** for a direction estimate (section 6.12).

If the signal disappears. A Wi-Fi network is followed on its own channel, read from the network's beacon (a strong network is also heard on the channels either side of its own, so the channel the radio happened to be on is not reliable). If the network has not been heard for about two seconds, the word under the number changes to **SEARCHING** in amber, the beeps stop, and the radio sweeps outwards from its channel (+1, -1, +2, -2 and so on, across channels 1 to 13) until the network is heard again. It then stays on that channel, and moves to the beacon's stated channel if that differs. If the network really has gone, it keeps searching and picks it up as soon as it comes back. You never need to leave and restart tracking. (Bluetooth and Zigbee targets do not need this: they are not channel-bound in the same way.)

> **TIP Getting a good reading**
> Turn slowly on the spot. Your body blocks the signal, so the reading is strongest when the device is in front of you. Signal strength jumps around indoors, so trust the trend over a few seconds, not one number. The signal on its own only tells you warmer or colder. **AIM** (section 6.12, beta) adds a direction estimate by measuring the signal while you turn.

The tracking screen also helps you find the spot.

- **Zoom.** The arc and the graph fill according to the last 10 seconds of signal instead of the whole -100 to -30 dBm range, so small changes show when you are close. The small word under the arc says **zoom** or **full**; **tap the arc** to switch.
- **HOTTER / COLDER.** Compares the last second with about three seconds ago.
- **Peak mark.** A white tick on the arc shows the strongest signal so far. It is held, and relaxes slowly if you do not find it again.
- **PASSED IT.** If the signal falls 6 dB or more from the peak soon after you were near it, the screen says **PASSED IT / turn back** in amber and plays a falling two-tone cue once. It clears when you come back near the peak. It is not used for very weak signals, and QUIET silences the cue.



Getting hotter | At the peak | Walked past it

6.4 Hide a device you do not care about, and bring it back

Your own cameras, or a neighbour's doorbell, can fill the list. **Long-press a row** for about a second: the toast "Hidden: <name>" appears and the device is removed from the list. It will not be shown, counted, alerted on or logged again, even after a restart. Up to 32 devices can be hidden (the oldest is dropped after that).

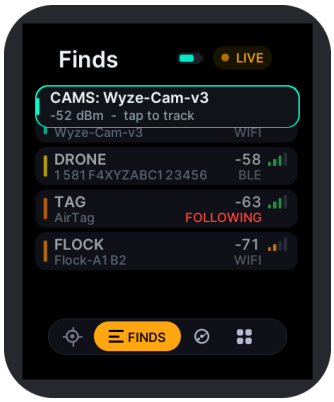
Nothing is lost. Hidden devices are kept by name on the **Ignored** screen, and there are three easy ways to reach it:

- On the **FINDS** page, tap **N hidden** next to the count at the top (shown when there is room), or the **N hidden - tap to manage** row at the end of the list.
- In **SETUP**, tap **Ignore list**.
- On the Ignored screen, tap a device to bring it back, or **RESTORE ALL**. A restored device shows up again the next time it is heard.

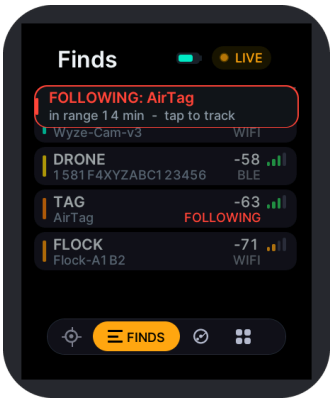
> **TIP Testing with your own network**
> If you hid your home network by accident, tap **N hidden** on the FINDS page and tap it in the list. Each hidden device is shown with its name, type and address so you can recognise it.

6.5 Alerts: banners, chimes and the find alert

- A **banner** slides in under the header for about five seconds with the group, the device name and its signal. Tap it to start tracking. A strong (high-confidence) find also wakes a sleeping screen.
- Each group has its own three-note **chime**, so you can tell what was found without looking.
- **SETUP > Find alert** chooses how often this happens: **Off**, **New group** (the default: one alert the first time each armed group finds something), or **Every find**. There is always a gap between alerts (at least four seconds, or 1.5 seconds in Every find mode), and the setting plays a sample when you change it.



Find banner



Tracker alert

6.6 Tracker alert ("following you")

A tracker that stays near you for a long time is worth noticing. When a device in the **TAGS** group has been continuously in range for the chosen time, you get a red **FOLLOWING** banner, an urgent alarm and a red tag in the list. Choose **SETUP > Tracker alert**: **5**, **10** (default) or **20** minutes, or Off. "In range" means it was heard within the last 90 seconds. It alerts once per tracker. It measures continuous time in range, so a tracker that keeps leaving and coming back may not trigger it.

6.7 Pause, quiet and power saving

Feature	How	What it does
Pause	Tap the status pill (or give three hard shakes if Shake to pause is on)	Stops recording new finds. Tap again to resume.
Discreet mode	Long-press the pill, or SETUP > Discreet mode	Silent: no sounds, no banners, no screen wake. Screen dims after 8 seconds and turns off after 20. Brightness is capped low. Scanning and logging continue.
Power saver	SETUP > Power saver: Off / Auto / On	Screen dims after 10 seconds and turns off after 30, brightness is capped, Bluetooth scanning and Wi-Fi channel changes slow down. Auto turns it on at 15% battery or less.
Low-battery banner	Automatic	Warns at 20% and again at 10%, unless the device is charging.
Screen sleep	SETUP > Auto-dim (off by default)	When switched on: dim after 30 seconds, off after 2 minutes. A touch wakes it (the first touch only wakes). Power saver and Discreet mode dim the screen on their own, whatever this is set to.

6.8 Turn the screen (orientation)

SETUP > Orientation cycles through **Portrait**, **Landscape**, **Portrait flip**, **Landscape flip** and **Auto**. Landscape shows the tiles as a wide three-by-three grid. The choice is remembered.

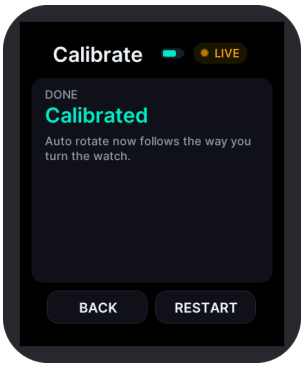
1 For **Auto** (follows how you hold the watch), choose Auto and then tap **Calibrate Auto** once.

2. **Step 1:** hold the board upright, the way the screen reads normally, and keep still until the bar fills (about 1.5 seconds).
3. **Step 2:** lay it flat on a table with the screen facing up and keep still until the bar fills.
4. "Calibrated" appears and the app returns to Setup. The result is saved. Run it again any time from the same row.
5. The screen only turns when the tilt is clear and steady for about 0.7 seconds, and it ignores the board lying flat, so it does not flicker.

> **NOTE Why calibrate?**
> The board's motion sensor can point in different directions relative to the screen and this is not documented. The calibration learns it for your board. It also makes **Raise to wake** use the correct axis. "Auto angle" is an extra manual quarter-turn fine-tune if you ever need it.



Calibration step 1



Done



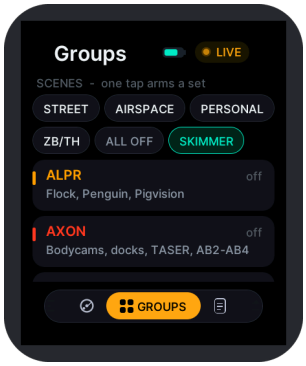
Landscape ARM page

6.9 Skimmer watch for gas pumps and ATMs

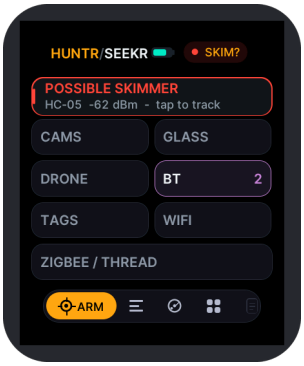
Some card skimmers are hidden inside gas pumps or ATM card readers and send the stolen card data to a thief over Bluetooth, using a cheap serial Bluetooth module that still carries its factory name. The WATCH listens for those names and warns you loudly if it hears one.

1. Walk up to the pump or machine and open **GROUPS**, then tap the **SKIMMER** scene chip. (It arms Bluetooth and switches the Skimmer alert on. Any armed group that listens by Bluetooth works too, and the alert does not need the BT group to be armed.)
2. Stay near the pump for 10 to 20 seconds. A suspect has to be heard several times, so one stray signal does not set it off.
3. If a suspect is heard you get the warning described below. **Do not use that pump or machine.** Tap the red banner to track the signal and see which way it gets stronger.

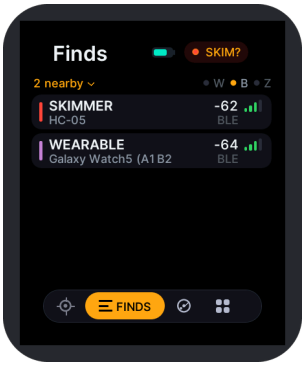
What happens	Details
Red banner	**POSSIBLE SKIMMER**, with the device name and its signal. Tap it to start tracking.
Siren	A two-tone siren, about 2.5 seconds. It follows the Volume setting and is silent in Discreet mode.
Screen	A sleeping or dimmed screen wakes up.
Status pill	Turns red and says **SKIM?** for 3 minutes after the last time the device was heard.
Repeats	While the device stays in range the warning repeats about once a minute, at most four times.
Record	The device appears in FINDS with a red **SKIMMER** tag and is written into the findings record.



SKIMMER scene chip



The warning



In the FINDS list

IMPORTANT Know the limits
This only sees Bluetooth Low Energy. The most common gas-pump skimmers use *Classic* Bluetooth modules (HC-05 and HC-06), and the WATCH has no Classic Bluetooth, so it cannot see them at all. It can catch skimmers built on a Bluetooth LE module (for example an HC-08). **A quiet screen does not mean a pump is safe.** Skimmers that store the data inside or send it by other means are invisible to this feature too.

NOTE A warning is a suspicion, not proof
It matches names only. Many harmless products and hobby projects use the same modules, and a thief can rename a module. The names watched for are HC-03, HC-05, HC-06, HC-08 (also written HC05 and HC06), linvor, and RNBT-xxxx with four hex characters; the match is exact, and capital letters do not matter.

TIP If you get a warning
Do not put your card in. Move to a different pump, preferably one in the attendant's view, or pay inside. Look for signs of tampering (a loose card reader, broken security seals). Tell the attendant or the bank, and report it to your local authorities. If it keeps warning about a device you know is harmless, long-press its row in FINDS to hide it. To switch the feature off use **SETUP > Skimmer alert**.

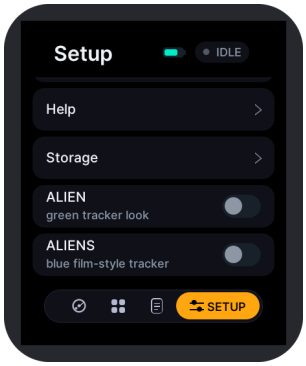
6.10 Side buttons, sleep and restart

The watch has two side buttons. **PWR** is connected to the power chip, so its presses are read from the chip. **BOOT** is an ordinary button once the watch is running. **To switch the watch off completely, hold PWR for 6 seconds.** The power chip does that in hardware, whatever the software is doing. Press PWR once to turn it on again.

Button	What it does
PWR, short press	Turns the **screen off** (scanning carries on) or back on.
PWR, held about 1 second	Opens the **Power screen** (below), waking the screen first if it was off. (Keep holding to 6 seconds and the power chip switches the watch off instead.)
PWR, held 6 seconds	The power chip switches the whole watch off. Press PWR again to turn it on.
BOOT, short press	Shows the **watch face** (section 6.11); press again to go back. While you are tracking a device, calibrating the touch or sharing, it turns the screen off or on instead, so you are not interrupted.
BOOT, held about 1.5 seconds	The screen says **Release to sleep**. Let go, and a short countdown starts: **sleep** unless you touch the screen.

SETUP > Power (near the bottom of the list), or a flick left past the last page, opens the Power screen as well. It has two large buttons. Tap one and a 2.5 second countdown starts; **touching the screen at any time cancels it**, and so does BACK.

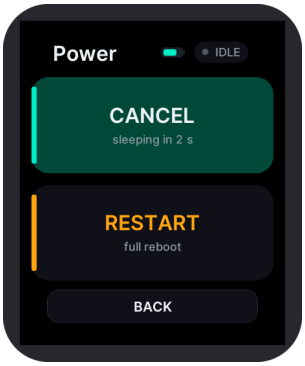
Power screen button	What happens
SLEEP	The radios, the amplifier and the display are switched off and the processor idles at half speed (a plain **standby**). **Your findings are kept**: when you wake it, scanning carries on where it stopped. Wake it with **BOOT**, **PWR** or by touching the screen.
RESTART	A complete reboot. Settings are kept; the unsaved findings are not.



SETUP > Power



The Power screen



Counting down

Which one to use	Screen off	Sleep (standby)
Scanning	Carries on	Stops
Findings	Kept	Kept
Battery use	Lower (the screen is the biggest user)	Lower again, but the processor is still awake, so it is far from zero
Wake with	Short PWR press, or a touch (BOOT wakes it onto the watch face)	BOOT, PWR or a touch

Battery protection	Details
Warnings	A banner at 20% and again at 10%.
Empty	At **3% or less**, not on USB, for three readings in a row (about 10 seconds), the watch saves the log and says **Battery empty - log saved, sleeping**; after a 5 second countdown that a touch cancels, it goes to sleep. The power chip's own cut-off protects the cell. USB connected: it never does this.

TIP Check the buttons
Open **SETUP > Self-test**: **PWR key** shows the PWR counts (from the power chip, and from the pin if that is switched on) and **BOOT key** shows BOOT (up or DOWN, and how many presses).

NOTE If the watch gets slow or jerky
Open **SETUP > Self-test > Speed** and note the three numbers (the last frame, the worst frame and the longest pause in the last 5 seconds), then restart the watch from the Power screen (RESTART) and mention what you were doing when it started in a bug report. A restart always clears it.

6.11 The watch face

Press **BOOT** briefly and the watch shows an **analog watch face** with the time from its clock chip. Press BOOT again, or swipe right, and you are back on the page you were on. Scanning carries on behind the face, and a find or tracker alert still slides in over it.



The default face | 3:00 | Clock approximate

On the face	What it shows
Dial	12 hour batons (the 12 o'clock one in amber), 48 minute marks, and the numerals 12, 3, 6 and 9.
Hands	The hour hand creeps smoothly between the numbers, the minute hand moves with the seconds, and a thin amber second hand steps once a second.
Centre	The **boot logo**: the radar mark with HUNTR/SEEKR above it. If you chose your own picture (SETUP > Boot logo > Custom) it appears there instead, at half brightness in a round window (a large picture is scaled down, and the wordmark is left out, as on the boot screen).
6 o'clock	The day and date, for example SUN 4 OCT.
Top	The battery level, and LIVE while the watch is scanning.

BOOT does this	When
Shows the face; the next press returns	From the main pages, the Self-test and the Power screen. You return to the same screen, on the same page.
Wakes the display onto the face	When the display is off or dimmed.
Keeps its old job (screen off or on)	While you are tracking a device, calibrating the touch, or sharing from the phone page, so none of those is interrupted.
Starts a sleep countdown	Held for 1.5 seconds, as before.

The time is read from the watch's clock chip (the PCF85063) several times a second, so the second hand steps exactly with the chip. To set the clock, open the **Live page on your phone** (SHARE): the page sets it. A watch whose clock has never been set starts from the build time and marks the date **approx** in amber until you do. If the chip cannot be read, the face shows the classic **10:10** and **clock not set**. The dial is a 12-hour dial, and the day and month are shown in English.

> **IMPORTANT Auto-dim is off by default, so the face stays lit**
> Like every screen, the face stays on until you switch **SETUP > Auto-dim** on (or use the Power saver or Discreet mode), which dims it and then switches the display off when idle; a touch or a button wakes it. A lit AMOLED uses battery, and a still dial left on for hours can leave a faint ghost image, so Auto-dim is worth turning on if you use the face a lot.

> **NOTE Screen off and on**
> A short press of **PWR** turns the screen off and on, and the face is still there when the screen comes back.

6.12 Find the direction: AIM (beta)

The watch has one antenna, so it cannot measure a direction directly. **AIM** works it out another way: you turn on the spot while the watch records how strong the signal is at each heading, using the watch's built-in **gyro** to know how far you have turned. The side where the signal was strongest is the estimate for where the device is. It is an **estimate, not a measurement**: your body, walls and reflections all bend the result.

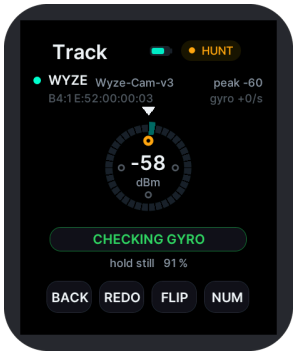
How to do it

1. Track a device as usual (FINDS, select it, TRACK) and stand in the open. Pick a landmark straight ahead. Hold the watch the way you will keep it for the whole routine, for example forearm across your chest with the screen facing you.
2. Tap **AIM**. First comes a **gyro check**: hold still for 2 seconds (**CHECKING GYRO**, then **GYRO READY**). It measures the gyro's own offset so slow drift does not creep into the heading.
3. **Point 1:** keep still, facing where you are. The pill says **HOLD STILL** while it records for about 3 seconds (it needs a few packets from the device).
4. **Points 2 to 4:** turn **RIGHT** (clockwise) about 90, 180 and 270 degrees. The pill counts down (**TURN RIGHT 50°**); when the amber dot is under the white triangle, hold still about 3 seconds until the dot turns green.
5. **Point 5:** turn back to where you started. You should be facing your landmark again: that is your check that the turning was measured well.

6 The watch now shows an **amber arrow** (the estimated direction) and an **amber arc** (how unsure it is). The pill says **TURN LEFT / RIGHT n°**; turn until it says **AHEAD** and walk that way, checking the number on the Track screen (**NUM**).



Track: tap AIM.



Gyro check. Hold still.



Point 1. HOLD STILL while it records.



Turning. The pill counts the degrees left.



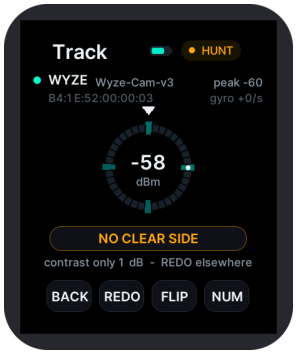
Point 5. Back to the start.



Calibrated. Arrow, arc and the turn needed.



AHEAD. The device is estimated in front of you.



No clear side. Redo, somewhere calmer.

What the watch checks for you

Situation	What the watch does
You move during the gyro check	**HOLD STILL: moving**. The 2 second window restarts. A small hand tremor is allowed; a steady turn is not.
You turn the wrong way	At point 2 the watch notices a clear turn LEFT, swaps the gyro direction by itself and says *Gyro direction swapped: turn RIGHT*. If you overshoot, the pill says TURN LEFT n°.
You move while a point records	That point restarts when you turn quickly or drift more than 30° off it.
Too few packets at point 1	It waits: *waiting for packets (n of 6)*.
The signal vanishes at points 2 to 5 (your body is in the way)	After about 6 seconds the point is saved with the missing packets counted as very low: **NO SIGNAL / body in the way**. The watch stays on the network's channel instead of hopping.
Points 1 and 5 read very differently	**NO CLEAR SIDE - readings vary**. Tap REDO.
You turn too fast (over about 90°/s)	**TOO FAST - slow down**. The gyro still counts, but no signal is recorded while you turn.
The motion sensor does not answer	**GYRO NOT WORKING**. See section 11.

Reading the screen

On screen	Meaning
White triangle at the top	The way you are facing now. The ring turns under it as you turn.
Dots inside the ring	The calibration points. Hollow = later, amber = do this one now, green = recorded.
Coloured segments	The average signal in each 10° of heading (teal = weaker, amber = stronger).
Amber arrow and arc	The estimated direction and how unsure it is (the ± in the pill, never better than ±15°). The white dot marks the single strongest segment.
Pill	Instructions while checking and calibrating; afterwards TURN LEFT / RIGHT n° or AHEAD, or NO CLEAR SIDE.
gyro +NN/s (top right)	Your turn speed in degrees per second, clockwise positive. About 0 when still; about +30 to +40 at a good turning pace.
BACK / REDO / FLIP / NUM	BACK leaves Track. **REDO** runs the gyro check and calibration again. **FLIP** reverses the gyro direction by hand (normally not needed). **NUM** returns to the dBm number (the gyro switches off).

TIP For a tighter result
After calibration the watch keeps recording the signal against your heading whenever you turn. Once most of the ring has data, the arrow comes from those 36 segments instead of the five points. Turn slowly (about 30°/s) on the spot for a lap or two, then hold still.

IMPORTANT Know the limits
AIM is a beta feature and an estimate. The main error is the signal itself changing between points (reflections, people moving, a device that is itself moving), which no gyro can fix. Indoors, expect a rough direction, not a bearing you can bet on; use the warmer / colder number to confirm. After a long walk with the arm swinging the heading can drift: tap REDO. Results are hints, never proof of where a device is or what it does.

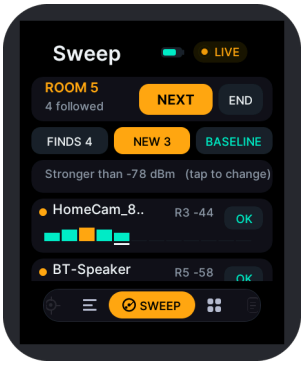
NOTE Reporting a problem with AIM
Note what the screen said (for example GYRO NOT WORKING, NO SIGNAL / body in the way, or the calibration message), whether you were walking or standing, and how you held the watch. Include that in your report.

6.13 Sweep: rooms, baseline and new devices

The **SWEEP** page is the third page of the menu (the radar icon). It helps you search a place systematically, with two tools on one page: **walkthrough marks**, which tell you in which room each device was strongest, and a **baseline**, which remembers what is normally in a place so you can see only what is *new*. The room control and the three tabs (FINDS, NEW, BASELINE) stay at the top while the list under them scrolls.



FINDS tab: ranked by room



NEW tab: not in the baseline



BASELINE tab: learning

Walkthrough marks

1. Arm the groups you are interested in on the ARM page (for example CAMS and BT). The walkthrough follows what those groups find.
2. Open the **SWEEP** page and tap **START**. You are in ROOM 1.
3. Walk the space. Tap **NEXT** each time you move into the next room (up to 12 rooms, and up to 32 devices are followed).
4. On the **FINDS** tab each device shows one small bar per room. The strongest room is amber and your current room is underlined; the best room and its signal are written on the right. Tap a device to track it; going back returns to this page.
5. Tap **END** to stop recording. **SAVE results** (at the bottom of the list) writes a CSV file with every device's strongest signal in each room.

> **TIP Read it as a guide, not proof**
> Signals are smoothed, and a strong device can be loud in two neighbouring rooms. The room where it peaks is where to look first; then use the tracking screen.

Baseline, then sweep

1. Stand in the place when it is **normal** (everything as it usually is). On the BASELINE tab tap **LEARN**. The time button cycles 3, 5 or 10 minutes; longer hears more. The radios run even if no group is armed.
2. When the time is up (or you tap STOP) the baseline is saved to the watch's flash. It survives restarts.
3. Later, tap **COMPARE** and walk the place. The **NEW** tab lists devices that were not in the baseline: heard at least three times, stronger than the threshold (tap the dBm figure to change it) and not on your ignore list. If a new device turns out to be fine, tap **OK** to add it to the baseline.

> **IMPORTANT What the baseline cannot do**
> It can only recognise devices whose address stays the same. Phones and many wearables use private Bluetooth or Wi-Fi addresses that change every few minutes, so they are skipped on purpose (otherwise every phone would look new). The table holds 716 devices. A device that was switched off while you learned will look new.

What	Where it is kept	Kept over a restart?
The baseline	/baseline.bin on the watch's flash	Yes
Walkthrough results (CSV)	/sweep_ followed by the time, on the flash, when you tap SAVE results	Yes
The rooms, the follow list and the new-device list	In memory	No

7. Drone Remote ID

Many drones are required to broadcast **Remote ID**: a short message giving the drone's serial number, its position, height and speed, and often the pilot's position. It travels over Bluetooth or Wi-Fi and anyone can read it. HUNTR/SEEKR reads it when the **DRONE** group is armed.

- The drone's **serial number** becomes its name in the list.
- On the tracking screen a card shows the ID, the drone's latitude and longitude, its height and speed, and the **pilot's position** when it is broadcast.
- The position and pilot position are also written to the saved log and shown on the phone Live view.
- Signal strength still works as usual for finding the drone; the reported position is what the drone says about itself, and it may be missing or wrong.



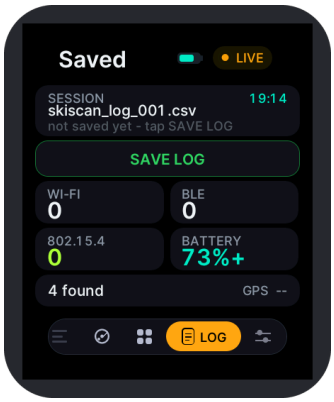
Tracking a drone with Remote ID

> **NOTE Nothing here gives you a compass arrow**
> The device has no GPS or compass, so it cannot point to a position. You can copy the coordinates into a maps app on your phone.

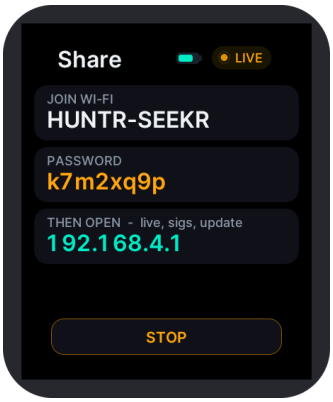
8. Sharing over Wi-Fi with your phone

The board can create its own Wi-Fi network so your phone can download logs, see a live list, edit signatures, set the clock and update the firmware. It needs no internet.

1. On the board go to **LOG** and tap **SHARE**. The screen shows a network name that starts with **HUNTR-SEEKR-** (four more characters, new at each Share), a random **password** and the address **192.168.4.1**. The password changes every time.
2. On your phone open Wi-Fi settings and join the network named on the board (**HUNTR-SEEKR-xxxx**) with that password (up to two devices can join). If the phone warns there is no internet, stay connected.
3. Open a browser on the phone and go to **http://192.168.4.1**.
4. When you have finished, tap **STOP** on the board. Scanning resumes.



LOG page: SHARE and REPORT



Share screen

Web page	What you can do there
Home (192.168.4.1)	List, download and delete the saved log files ("Wipe all logs" asks first), and open the pages below.
Live view (/live)	A list of current finds that refreshes every 1.5 seconds, with signal, tracker flags and drone positions. Opening it also **sets the board's clock from your phone**.
Signatures (/sigs)	Add your own detection rules (section 9).
Update (/update)	Install new firmware from a file (section 9).

> **IMPORTANT While the hotspot is on**
> Wi-Fi scanning pauses because the radio is busy running the hotspot. Bluetooth scanning keeps going and everything already found stays in the list. Stop the hotspot to return to full scanning.

9. Logs, signatures, clock and updates

9.1 The findings record: save what was found, when you choose

The board keeps a **findings record** of the session in memory: every network and device it has found, one entry each, with when it was first and last heard, its best signal, channel, how many times it was heard, and where you were if a GPS is fitted. **Nothing is written to flash while you scan.** You decide when to save it, so scanning can never quietly fill the storage.

- **SAVE LOG** on the LOG page writes the record to one file, for example **skiscan_log_004.csv**. Tap it again later and the *same file is rewritten* with the new finds: it never grows by repetition.
- The LOG page's session card says exactly what is saved ("saved 21:15 - 4 devices"), or "not saved yet".
- The record holds up to 192 devices; if more are found, the ones heard longest ago make way and the toast says how many were left out. A saved file is a few tens of kilobytes at most.
- A device you have hidden (section 6.4) is not recorded. If less than 40 KB of storage is free, SAVE LOG refuses and says so instead of filling it.

SETUP > Save log chooses how saving works:

Setting	What it does
Manual (default)	Saves only when you tap SAVE LOG. Starting a new session with unsaved finds shows a reminder; nothing is written behind your back.
Auto 5 min	Refreshes the same file every 5 minutes if something changed, and whenever you stop or pause scanning. Silent unless it fails.
Live	The old way: every new find is added to a separate file (**skiscan_live_NNN.csv**) the moment it is found, once per device per session, up to 1500 rows, stopping when less than 64 KB is free. Tracking trails (**skiscan_hunt_NNN.csv**) are written only in this mode and only with a GPS fix.

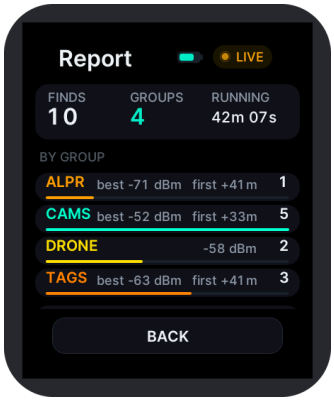
Files are created only when something is saved, so starting and stopping sessions leaves no empty files behind. Get the files from the phone page (section 8). Each is a plain text table that opens in any spreadsheet. To free space, **SETUP > Storage > DELETE LOGS** (tap twice) removes the saved log files and leaves your settings, ignore list and signatures alone.

Column	Meaning
first_seen, last_seen	Date and time from the board's clock, or +HH:MM:SS from the start of the session if the clock is not set.
name	The device's network name (SSID) or ID.
mac	Its hardware address.
family, type	What it matched (for example WYZE, TAG, DRONE) and how: by hardware address (OUI), network name (SSID), Bluetooth name or type tag (WEARABLE, LIGHT...), or a Remote ID broadcast.
radio, channel	How it was heard (WIFI, BLE or ZB) and on which channel.
best_rssi, last_rssi, hits	Strongest and latest signal in dBm, and how many times it was heard.
confidence	LOW (just a nearby network or device), MED or HIGH (a strong match).

Column	Meaning
lat, lon	Where you were when it was first heard. Blank unless a GPS is fitted and has a fix.
rid_lat, rid_lon, rid_alt	The drone's own reported position and height, from Remote ID.

If a GPS module is fitted and at least one find has a position, SAVE LOG also writes the same record as a map file next to the CSV, with the same name ending **.kml** and **.geojson** (for example **skiscan_log_004.kml**). Open either in Google Earth, QGIS or geojson.io to see where each find was heard. A drone's own reported position is added as a second point.

LOG > REPORT shows a summary of the current session on the board: total finds, the number of active groups, the running time, a bar for each group with its best signal and when it first appeared, and a chart of finds per minute over the last hour.



Session report



Finds per minute, last 60 minutes

9.2 The clock

A small battery-backed clock stamps the log. The time shows at the top right of the LOG page's session card: teal when set from your phone, **amber** when it is only an estimate from when the software was built. To set it exactly, open the Live view (section 8) once from your phone; that sets the board's clock from the phone's clock and time zone.

9.3 Your own detection rules (signatures)

Open **192.168.4.1/sigs** on your phone while connected to the hotspot, type rules one per line, and press Save. They are used in addition to the built-in lists and are kept after a restart. Up to 128 rules. A line that starts with # is a comment.

Rule	Example	Matches
oui	oui B4:1E:52 ALPR high	Any device whose hardware address starts with those three bytes (Wi-Fi or Bluetooth).
ssid	ssid MyCamNet CAMS	A Wi-Fi network name containing the text (not case sensitive, at least 3 letters).
ble	ble MyTag TAGS	A Bluetooth device name containing the text.
cid	cid 0x0171 GLASS	A Bluetooth manufacturer (company) ID.
ie	ie 3A91C0F2 CAMS	A device fingerprint code (eight hex digits) as shown in a Wi-Fi find, such as IE-fp 3A91C0F2 (section 5.2).
uuid	uuid 0xFE65 TAGS	A Bluetooth service ID (a 16-bit number, as four hex digits) that the device advertises.

After the word or code comes the **group** it should go in (ALPR, AXON, CAMS, GLASS, DRONE, BT, TAGS, WIFI, or a camera brand such as RING or WYZE) and, optionally, **low**, **med** or **high** confidence. The page shows how many rules loaded and how many lines were ignored as invalid.

Signature updates. A second file named **sigs_update.txt** uses the same format and is read after your own rules, so a list you receive from someone else can be added without touching your own. Upload it from the hotspot's home page (192.168.4.1, the file chooser there): it takes effect at the next start-up. Keep your own rules in /sigs and use sigs_update.txt only for lists you receive.

9.4 Update the firmware from your phone

1. Download **HUNTR-SEEKR_Watch_Public_Beta_1_app-only.bin** from the project's Releases page to your phone.
2. Connect USB power, or make sure the battery is above 30%. Then tap **SHARE** on the LOG page, join the hotspot and open **192.168.4.1/update**.
3. Choose the .bin file and press **Install**. Do not remove power. The watch restarts by itself when it is done.

> **IMPORTANT Update safety**
> The page refuses on low battery, refuses files that are too small to be real firmware, and only restarts if the new image passes the final check. Even so, keep the watch powered during an update. If anything ever goes wrong you can always flash the full .bin file over USB (section 3.2).

> **NOTE Which file?**
> The **app-only** file is for this phone update. The complete **HUNTR-SEEKR_Watch_Public_Beta_1.bin** is for flashing over USB with esptool (section 3.2); do not choose it on the update page.

9.5 A boot sound of your own

The WATCH has a real audio chip, so it can play a short sound of your choosing on the boot screen instead of the built-in chime.

1. Press **SHARE** on the LOG page, join the **HUNTR-SEEKR-xxxx** hotspot (the name shown on the board) and open **192.168.4.1**, then tap **Boot sound** (or go straight to **192.168.4.1/boot**).

2. Tap the file picker and choose any short audio file: MP3, WAV, M4A, OGG and so on. The page converts it in your browser, so you do not need to prepare it. It keeps the **first 6 seconds**, makes it mono at 16 kHz, evens out the level and fades the last 40 ms. You can listen to the result on the page before sending it.
3. Press **Upload to the watch**. The page says **OK - saved 3.0 s** and the boot sound is switched to **Custom** for you. **Play it on the watch** tries it straight away.
4. To change what plays at start-up, use **SETUP > Boot sound**: **Off**, **Chime** (the built-in tune, the default) or **Custom** (only offered once a sound is stored). Choosing Chime or Custom plays a preview.

Item	Detail
File format on the watch	WAV, uncompressed PCM, mono, 16 kHz, 16-bit, at most 6 seconds (192 KB). The page produces exactly this. If you upload a WAV yourself it must already be in this format with the standard 44-byte header; anything else is refused and the page says why.
Size	32 KB per second: 3 seconds is about 96 KB. It is streamed from flash in small pieces, so length does not use memory. Uploads are refused if less than 40 KB of storage would be left free; delete old logs first (SETUP > Storage > DELETE LOGS).
During boot	A long sound keeps the boot screen up until it has finished (up to 6 seconds). Tap the screen to skip: the sound fades out at once.
Volume and quiet modes	It follows the Volume setting and Discreet mode, like every other sound: volume 0 or Discreet means silence.
If something goes wrong	If the file is missing or damaged, the built-in chime plays instead of silence. Delete on the Boot sound page removes the file and returns to the chime.
Housekeeping	DELETE LOGS and the web page's Wipe all logs never touch the boot sound. It is stored as **bootsound.wav**, which you can download from the hotspot's home page as a backup.

9.6 A boot logo of your own

The boot screen normally shows the radar rings and the HUNTR/SEEKR title. You can replace those with your own picture. The progress bar and the version line stay.

1. Press **SHARE** on the LOG page, join the **HUNTR-SEEKR-xxxx** hotspot (the name shown on the board) and open **192.168.4.1/bootlogo** (or tap **Boot logo** on the home page).
2. Choose any picture: PNG, JPG, BMP, GIF or WebP. The page converts it in your browser: it shrinks it to fit this screen (the page states the largest size, for example 280 x 280 px on the WATCH), turns transparent areas black, and shows you the result.
3. Press **Upload to the device**. The page says **OK - saved 280 x 140** and the boot logo is switched to **Custom** for you. **Show it on the device** displays the boot screen straight away.
4. **SETUP > Boot logo** switches between **Default** and **Custom** (only offered once a picture is stored). Each change shows a preview of the boot screen.

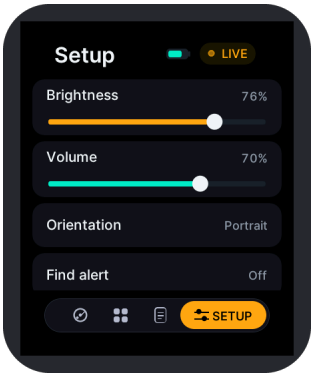
Item	Detail
Largest picture	280 x 280 pixels. The page makes the picture fit for you. A picture that is taller than the room on a given screen (for example in landscape) is shrunk to fit when it is drawn, keeping its shape; it is never enlarged.
Your existing boot.bmp	A plain **24-bit uncompressed BMP** uploaded from the hotspot's home page also works as it is, up to 1024 x 1024 (it is shrunk to fit when drawn). If both exist, the picture from the Boot logo page is used. 8-bit, compressed and other BMP types are not supported: use the Boot logo page and it will convert them.
Size on flash	A 200 x 200 picture is 78 KB. It is read from flash while the boot screen is drawn, so it uses no memory. It is drawn once; afterwards only the progress bar is redrawn.
If something goes wrong	If the file is missing, damaged or too big for the screen, the normal radar boot screen is shown instead of a blank one. Delete on the Boot logo page removes the picture and returns to Default.
Housekeeping	DELETE LOGS and Wipe all logs never touch your picture. Uploads are refused if less than 40 KB of storage would be left free.

10. Settings reference

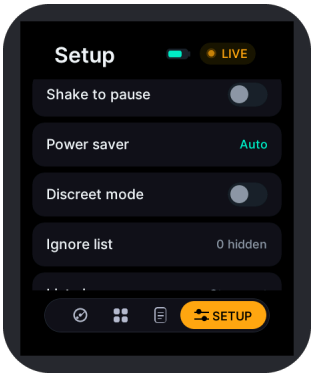
Find these on the **SETUP** page. Drag up to scroll; the list is long.

Setting	Options	Default	What it does
Brightness	Slider	About 78%	Screen brightness. The screen also dims automatically when idle.
Volume	Slider, 0 to 100%	70%	Loudness of all sounds. All the way left is silent. Plays a sample when you let go.
Orientation	Portrait, Landscape, Portrait flip, Landscape flip, Auto	Portrait	Which way the screen is drawn. Auto needs the one-time calibration.
Auto angle / Calibrate Auto	0, 90, 180, 270 / wizard	0 / not done	Only shown when Orientation is Auto. See section 6.8.
Find alert	Off, New group, Every find	New group	How often finds are announced by chime and banner.
Tracker alert	Off, 5, 10, 20 minutes	10 min	How long a tracker must stay near you before the FOLLOWING alert.
Power	Opens a screen	-	SLEEP and RESTART as two big buttons (section 6.10). Also reached by flicking left past the last page.
List view	Strongest, Strong only, Newest, A-Z	Strongest	How the FINDS list is ordered (tap the count above the list to change it too).
Save log	Manual, Auto 5 min, Live	Manual	How the findings record is saved (section 9.1).
Rogue Wi-Fi alert	On / Off	On	Announces DEAUTH, TWIN and KARMA signs (section 5.2) whichever groups are armed, as long as a Wi-Fi group is scanning.

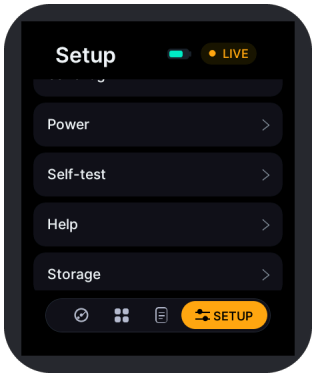
Setting	Options	Default	What it does
Bluetooth share	On / Off	Off	With Wi-Fi and Bluetooth groups armed together, pauses Wi-Fi for 2 seconds in every 6 so Bluetooth gets the whole radio (section 5.2).
Skimmer alert	On / Off	On	Warns when a Bluetooth LE module with a card-skimmer name is heard (section 6.9). Works whichever groups are armed, as long as scanning is running.
Auto-dim	On / Off	Off	When on, dims the screen after 30 s and switches it off after 2 minutes; a touch wakes it. Off by default, so the screen stays on. Turn it on to save battery and protect the AMOLED screen.
Raise to wake	On / Off	On	Wakes a dim or sleeping screen when you lift the board to look at it (only matters when Auto-dim, Power saver or Discreet mode is dimming the screen).
Shake to pause	On / Off	Off	Three hard shakes pause or resume scanning.
Power saver	Off, Auto, On	Auto	See section 6.7.
Discreet mode	On / Off	Off	Silent, dim, no banners. Same as a long-press on the status pill.
Boot logo	Default, Custom	Default	The picture on the boot screen. Custom is your own picture uploaded from the phone page (section 9.6).
Boot sound	Off, Chime, Custom	Chime	What plays on the boot screen. Custom is your own sound uploaded from the phone page (section 9.5).
Ignore list	Tap to open the Ignored screen	Empty	Shows how many devices are hidden. Tap a device there to restore it, or RESTORE ALL.
Self-test	Opens a screen	-	Checks every part of the hardware (next page).
Help	Opens a screen	-	A short in-device reminder of the groups and tips.
Storage	Opens a screen	-	Shows how full the flash memory is and lists the files on it.



Brightness, volume, orientation



Alerts and modes



Ignore list, Self-test, Help, Storage

11. Self-test and troubleshooting

11.1 The Self-test screen

SETUP > Self-test lists each part of the board with a green dot when it looks healthy and a red dot when it does not. It updates several times a second. **BEEP** plays the start-up melody so you can check the speaker.

Row	What green means	If it is red
Touch	The touch chip answers. Numbers show how many readings and touch signals it has sent.	Restart the board. If it stays red, flash the .bin again (section 3.2).
Display	Shows the screen size and brightness.	-
Sound	The audio chip (ES8311) was found. The text also counts the tap sounds sent to it and how many were lost (for example "212 blips, 0 lost").	Check the volume is above zero; flash the .bin again.
Motion	The motion sensor answers; live x/y/z numbers are shown.	Auto rotate and shake features will not work.
Clock	The clock has a valid time.	Open the Live view from your phone to set it ("approx" means estimated).
Battery	Shows percent, USB and charging.	"no battery detected" is normal on USB power alone.
Radios	Counts of Wi-Fi, Bluetooth and 802.15.4 signals received.	Zeros with a group armed: see the Wi-Fi row below. If the board restarted unexpectedly, this row shows **restarted in:** and the step it was in; note it if you report a problem.
Storage / Memory	Flash use, custom rules loaded, hidden devices; free memory and uptime.	Storage red means logs cannot be saved. Free space with SETUP > Storage > DELETE LOGS.
PWR key	The PWR counts: from the power chip.	Red: no PWR press has been seen. Press PWR and watch the chip counts.
BOOT key	BOOT: up, or DOWN while pressed, and how many presses.	Press BOOT and watch the count.

Row	What green means	If it is red
Speed	How fast the screen is drawing: the last frame time, the worst frame and the longest pause in the last 5 seconds. Green means a worst frame under 120 ms and a worst pause under 150 ms.	A page swipe will look jerky. Note the three numbers (they update every 5 seconds while you use the watch) and include them in a bug report.

11.2 Troubleshooting

What you see	What to try
Nothing shows on the screen	Check the cable is a data cable. Try another USB port. Flash the .bin again (section 3.2).
Touch does nothing	Boot screen says "touch NOT detected"? Unplug, wait, plug in again. Check Self-test > Touch. Flash the .bin again (section 3.2).
The first touch after a pause does nothing	Normal: that touch wakes the screen. Touch again.
No sound	Volume slider above zero? Discreet mode off (pill does not say QUIET)? Find alert not Off? Self-test > Sound should be green. Tap BEEP.
A tap sound is sometimes missing	Open SETUP > Self-test and look at the Sound row, which counts blips and lost sounds. **0 lost** while a tap was silent means the sound was sent but not heard: the amplifier was asleep after a pause (it now stays on for 10 seconds and wakes with the screen, so the first tap after a wake is covered). If the lost count goes up, include the numbers in a bug report. Also check the Speed row: a very long pause there is the audio waiting on a flash write.
AIM says GYRO NOT WORKING	The watch could not read its motion sensor (NO DATA or NO ANSWER). Open SETUP > Self-test > Motion: it should show three changing numbers. Restart the watch from the Power screen. If it stays red, AIM cannot be used on that unit; the normal Track screen is not affected.
AIM says NO CLEAR SIDE	The signal changed too little around you, or point 1 and point 5 disagreed. Move to open space (not next to a wall or metal), hold the watch the same way throughout, and tap REDO. A very weak or very steady signal gives no side to find.
The arrow seems to turn the wrong way	Tap FLIP once, then REDO. FLIP reverses the gyro's turn direction (the watch normally works this out by itself at point 2).
A calibration point keeps restarting	You moved while it was recording. Hold the watch still against your body for about 3 seconds at each point. A small hand tremor is allowed; walking, swaying or a quick shift is not.
The watch face says clock not set, or approx	The clock chip has not been set (**clock not set**, the face shows 10:10) or still holds the build time (**approx**). Open the **Live page on your phone** (SHARE): it sets the clock.
BOOT does not turn the screen off	A short BOOT press shows the watch face. PWR (a short press) turns the screen off and on.
The watch face never switches off	Auto-dim is off by default. Switch on **SETUP > Auto-dim** (or Power saver) and the face dims and the display goes off when idle.
PWR does nothing	Open **SETUP > Self-test > PWR key** and press PWR briefly: the chip's short count (**chip s**) should go up (hold for a second or so and **chip l** counts). If it never counts, include the numbers in a bug report. Holding PWR for 6 seconds always switches the watch off, in hardware.
The watch will not wake from sleep	Press BOOT, or press and hold PWR for a second or touch the screen. If it still does not wake, hold PWR for 6 seconds until it is off, then press PWR once to start it.

What you see	What to try
Bluetooth finds are slow to appear while a Wi-Fi group is armed	Turn **SETUP > Bluetooth share** on (section 5.2) so Wi-Fi pauses regularly and Bluetooth gets the whole radio. Arming only Bluetooth groups also gives Bluetooth the whole radio.
A DEAUTH, TWIN or KARMA warning appears	Section 5.2 explains each one. They are warnings, not proof; a venue with several access points can look like a TWIN. Tap the find to Track the source, hide it with the ignore list, or turn the alerts off with SETUP > Rogue Wi-Fi alert.
Wi-Fi groups find nothing	The list says "Wi-Fi radio: no frames yet" at first, then after 10 seconds "scanning access points". In that mode only access points are seen, in bursts. Wi-Fi also pauses while the Share hotspot is on. Make sure a Wi-Fi group (not ZB/TH) is armed.
A tile is dim red and says blocked	You armed groups from the other side of the radio rule (section 5). Switch the other side off first.
Screen turns the wrong way in Auto	Run **SETUP > Calibrate Auto** (section 6.8). If it is still off by a quarter turn, adjust Auto angle.
Log times are blank or amber	The clock is not set. Open the Live view from your phone once.
The phone will not stay on the hotspot	Some phones drop networks without internet. Turn off "auto-switch to mobile data" or similar while connected. The address is http://192.168.4.1 (not https).
A watch or phone shows only "Apple" or does not appear	Normal. Apple devices rotate their address and send very little, so they cannot be told apart from each other.
A light or watch shows as plain BT	Its name and maker did not match the built-in lists. You can add your own rule in /sigs (section 9.3).
Battery drains quickly	Turn Auto-dim on (it is off by default), turn Power saver On, lower Brightness, and use Discreet mode when you do not need the screen.

12. Care, safety and responsible use

- **It mostly listens.** It receives broadcasts. The access-point scan and Bluetooth scanning send the same small "who is there?" requests any phone sends. It does not jam, connect to, or interfere with other devices.
- **Know the law where you are.** Rules about scanning radio signals, recording device information and following people differ between countries and places. You are responsible for how you use it.
- **Results are not proof.** A match means a signal looked like a known kind of device. It cannot tell you whether a device is recording, who owns it, or why it is there. Names and addresses can be changed or randomized.
- **Do not use it to harass or confront anyone.** If a tracker alert worries you, move to a safe public place and contact the appropriate authorities or support services in your area.
- **Battery care.** Use only the battery type Waveshare specifies. Do not leave a charging battery unattended, cover it, or use a swollen or damaged battery.
- **Keep the software up to date** by installing newer releases.

Quick-reference card

Cut out or fold this page and keep it with the device.

MOVE Swipe left / right, or tap the bottom icons: ARM, FINDS, SWEEP, GROUPS, LOG, SETUP.	**ARM** ARM page: tap a tile. Or GROUPS page: tap a Scene chip.	**TRACK** FINDS: tap a device, then TRACK. Number closer to zero = closer. **AIM** = direction (6.12).
HIDE A DEVICE Long-press its row for one second.	**PAUSE / RESUME** Tap the pill at the top right.	**SILENT + DIM** Long-press the pill (it shows QUIET).
WI-FI vs ZB/TH Only one side at a time. Blocked tiles are dim red. BT and TAGS work with both.	**SIGNAL (dBm)** -30 to -50 very close -50 to -65 close -65 to -80 medium -80 to -90 far	**HOTSPOT** LOG > SHARE. Join HUNTR-SEEKR-xxxx (the name shown on the board), open http://192.168.4.1. Wi-Fi scanning pauses meanwhile.
SET THE CLOCK Open /live from your phone once while on the hotspot.	**AUTO ROTATE** SETUP > Orientation > Auto, then Calibrate Auto: upright, then flat.	**SOMETHING WRONG?** SETUP > Self-test. A red dot marks the problem.

Glossary

Term	Meaning
802.15.4	The low-power radio standard used by Zigbee and Thread smart-home devices. Separate from Wi-Fi and Bluetooth.
Access point (AP)	A device that provides a Wi-Fi network, such as a router or a camera that creates its own network.
Advertisement	A short Bluetooth message a device sends repeatedly to announce itself.
Beacon	The Wi-Fi equivalent: an access point announcing its name and details several times a second.
BLE	Bluetooth Low Energy, the power-saving kind of Bluetooth used by trackers and wearables.
Channel	A slice of the Wi-Fi band. The device steps through channels 1 to 13, listening on each briefly.
CSV	Comma-separated values: a plain text table that opens in any spreadsheet.
dBm	The unit of signal strength. Negative numbers; closer to zero is stronger.
Beacon types	iBeacon, Eddystone and AltBeacon are small, fixed Bluetooth messages that shops and venues broadcast to mark a place or object.
Fast Pair	Google's quick-pairing system for Android accessories such as earbuds.
Find Hub	Google's tracker network (formerly Find My Device). Its trackers broadcast a special Bluetooth frame that this device recognizes.
Hotspot	A Wi-Fi network created by the board itself (HUNTR-SEEKR-xxxx) so your phone can connect to it.
IP address (192.168.4.1)	The board's address on its own hotspot network. Type it into a phone browser.
MAC address	A device's hardware address, for example 2C:AA:8E:01:02:03.
OUI	The first three bytes of a MAC address; identify the manufacturer.
Remote ID	A broadcast from a drone giving its serial number, position and the pilot's position.
RSSI	Received signal strength indicator, the number shown in dBm.
SSID	The name of a Wi-Fi network.
Zigbee / Thread	Smart-home mesh networks (lights, sensors, locks) that run over 802.15.4.
Skimmer	A device hidden inside a card reader (gas pump, ATM) that copies card data. Some send it to the thief over Bluetooth.

Appendix A. Technical summary

Item	Detail
Board	Waveshare ESP32-C6-Touch-AMOLED-2.06
Display	2.06 inch AMOLED, 410 x 502 pixels, rounded corners, capacitive touch
Radios	2.4 GHz Wi-Fi 6, Bluetooth 5 (LE), 802.15.4 (Zigbee / Thread), one shared radio
Built-in extras	Speaker with audio chip, motion sensor (accelerometer and gyro), battery-backed clock, power-management chip with battery gauge
Storage	Logs and settings in on-board flash (about 9.9 MB available)
Hotspot	WPA2, name HUNTR-SEEKR-xxxx (four new characters at each Share), random 8-character password, address 192.168.4.1
Software	Firmware image for the ESP32. Public Beta 1